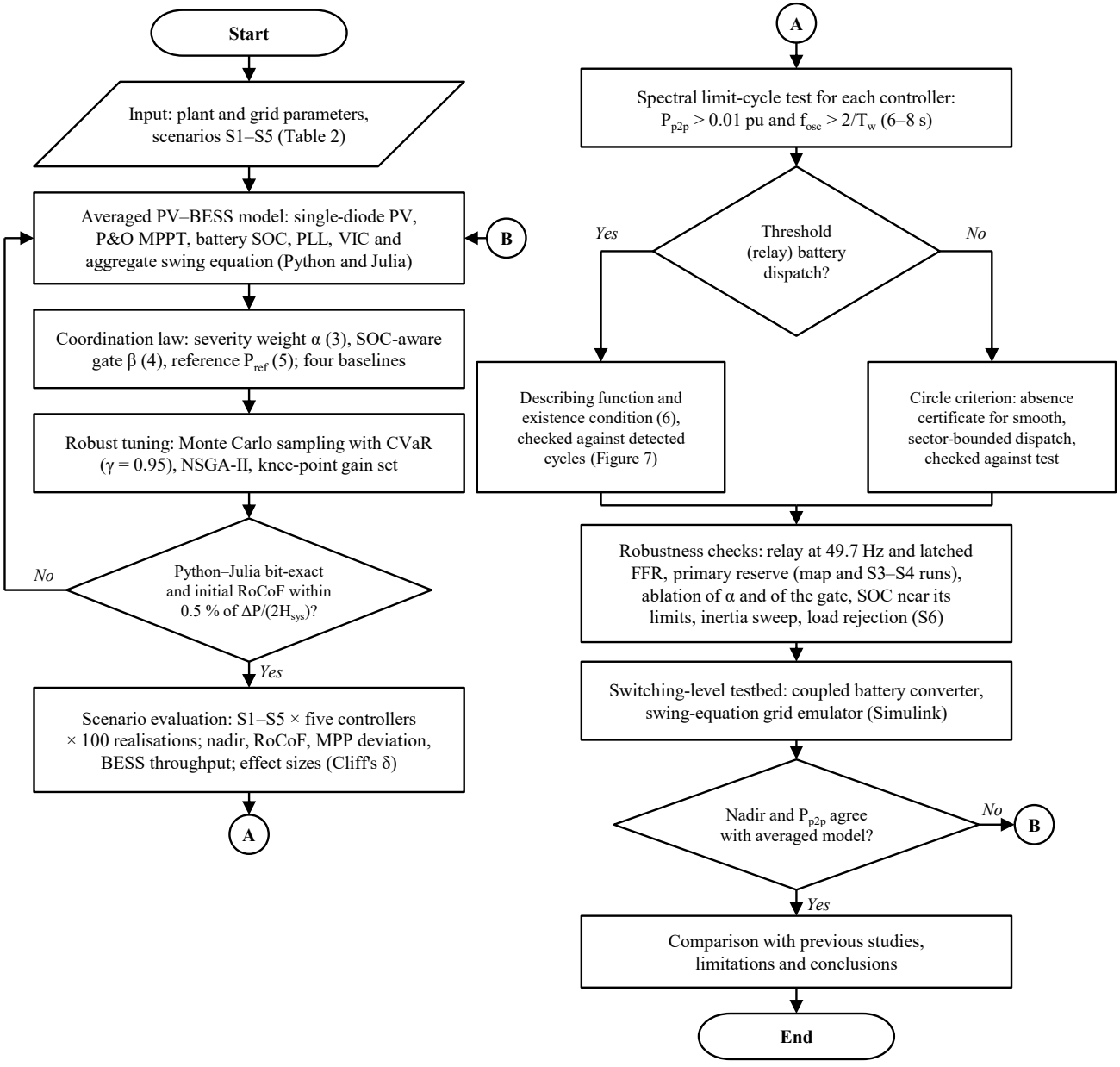

Start
Input: plant and grid parameters, scenarios S1–S5 (Table 2)
Averaged PV–BESS model: single-diode PV, P&O MPPT, battery SOC, PLL, VIC and aggregate swing equation (Python and Julia)
B
Coordination law: severity weight α (3), SOC-aware gate β (4), reference P_{ref} (5); four baselines
Robust tuning: Monte Carlo sampling with CVaR (γ = 0.95), NSGA-II, knee-point gain set
No
Python–Julia bit-exact and initial RoCoF within 0.5 % of ΔP/(2H_{sys})?
Yes
Scenario evaluation: S1–S5 × five controllers × 100 realisations; nadir, RoCoF, MPP deviation, BESS throughput; effect sizes (Cliff's δ)
A
A
Spectral limit-cycle test for each controller: P_{p2p} > 0.01 pu and f_{osc} > 2/T_w (6–8 s)
Yes
Threshold (relay) battery dispatch?
No
Describing function and existence condition (6), checked against detected cycles (Figure 7)
Circle criterion: absence certificate for smooth, sector-bounded dispatch, checked against test
Robustness checks: relay at 49.7 Hz and latched FFR, primary reserve (map and S3–S4 runs), ablation of α and of the gate, SOC near its limits, inertia sweep, load rejection (S6)
Switching-level testbed: coupled battery converter, swing-equation grid emulator (Simulink)
Nadir and P_{p2p} agree with averaged model?
No
B
Yes
Comparison with previous studies, limitations and conclusions
End